Spurious Finite-Time Modal Blowup in Low-Order Galerkin Approximations of Fast–Slow Stably Stratified Boundary Layers:

A PhD Research Project

PhD Candidate: *[Student Name]*
Supervisor: David England (UAH.Edu BLM)

August 29, 2026

1 Project overview

The stably stratified atmospheric boundary layer (SBL) exhibits fast–slow dynamics between turbulent kinetic energy (TKE) and mean shear, with strong timescale separation and sensitivity to stratification. Low-order Galerkin closures and single-column models can display spurious finite-time blowup or collapse of modal amplitudes, obscuring the physical interpretation of turbulence intermittency and regime transitions.

This PhD project will develop a rigorous fast–slow dynamical-systems framework for a prototypical SBL model, characterize its critical manifold and fold structure, and relate these features to numerical pathologies in low-order Galerkin approximations and operational boundary-layer schemes.

2 Core model and fast–slow structure

We consider a singularly perturbed system for turbulent kinetic energy E and mean shear S, with timescale separation $0 < \epsilon \ll 1$:

$$\epsilon \frac{dE}{dt} = f(E,S) = l\sqrt{E+\delta}\left(S^2 - \phi N^2 - c_b N^2 \frac{E}{E+\alpha}\right) - \frac{E^{3/2}}{l}, \tag{1}$$

$$\frac{dS}{dt} = g(E,S) = G - c_1 ES - c_2 S, \tag{2}$$

on the physical domain $\mathcal{D} = \{(E,S) \in \mathbb{R}^2 : E \geq 0\}$. Here l is the mixing length, N^2 the (constant) Brunt–Väisälä frequency, G the geostrophic forcing, $\phi > 0$ a stability-correction coefficient, $\delta, \alpha > 0$ regularization/saturation constants, $c_b = 0.50$, and $c_1 = 1.80\ \mathrm{s\,m^{-2}}$. The critical manifold is

$$\mathcal{C}_0 = \{(E,S) \in \mathcal{D} : f(E,S) = 0\}. \tag{3}$$

3 Research objectives

3.1 Objective 1: Well-posedness and regularity of the fast subsystem

- Establish dimensional consistency of $f(E,S)$ and $g(E,S)$, determining the required units of δ, α, ϕ, and c_2 so that f carries units of $\mathrm{m^2\,s^{-3}}$ and $E/(E+\alpha)$ is dimensionless, with $[G] = \mathrm{s^{-2}}$.

- Analyze C^1-regularity of f on $\mathcal{D}$, showing that $\delta > 0$ restores differentiability at $E = 0$ and yields a well-posed fast subsystem for geometric singular perturbation theory.

3.2 Objective 2: Critical manifold geometry and fold structure

- Derive the fast eigenvalue

$$\lambda_f(E, S) \equiv \frac{\partial f}{\partial E}(E, S), \tag{4}$$

  in closed form, including the product rule on $\sqrt{E + \delta}$, the quotient/chain rule on $E/(E + \alpha)$, and the $E^{3/2}/l$ term.

- Apply the implicit function theorem to show that at points $(E^*, S^*) \in \mathcal{C}_0$ with $\lambda_f(E^*, S^*) \neq 0$, the set $\mathcal{C}_0$ is locally the graph of a C^1 function $E = E^*(S)$, and characterize the loss of regularity at $\lambda_f = 0$.

- Classify branches of $\mathcal{C}_0$ into attracting ($\lambda_f < 0$), repelling ($\lambda_f > 0$), and fold ($\lambda_f = 0$) segments, using the fast subsystem $\epsilon \dot{E} = f(E, S)$ with S frozen to determine the timescale of attraction/repulsion.

3.3 Objective 3: Nondegenerate folds and slow passage

- Identify candidate fold points (E_0, S_0) satisfying $f(E_0, S_0) = 0$ and $f_E(E_0, S_0) = 0$, and compute $f_{EE}(E_0, S_0)$ and $f_S(E_0, S_0)$ explicitly.

- Formulate and verify nondegeneracy ($f_{EE} \neq 0$) and transversality ($f_S \neq 0$) conditions for generic saddle-node folds of the fast subsystem, interpreting each condition in terms of excluded degeneracies.

- For a nondegenerate fold with slow drift $\dot{S} \approx g(E_0, S_0) \equiv v_0 \neq 0$, approximate

$$f(E, S) \approx f_S(E_0, S_0)(S - S_0) + \tfrac{1}{2} f_{EE}(E_0, S_0)(E - E_0)^2,$$

  substitute $S - S_0 \approx v_0 t$ into $\epsilon \dot{E} = f(E, S)$, and derive the fractional scaling exponent relating the fold-crossing transient duration to ϵ (showing it is not $\mathcal{O}(\epsilon)$).

3.4 Objective 4: Reduced slow flow, hysteresis, and Galerkin blowup

- On attracting branches, derive the reduced slow dynamics

$$\frac{dS}{dt} = g(E^*(S), S),$$

  where $E^*(S)$ solves $f(E, S) = 0$ on the attracting portion of $\mathcal{C}_0$.

- Analyze multivalued (S-shaped) $E^*(S)$ in the presence of fold pairs, and show how slow variation of S or forcing G produces hysteresis in E, corresponding to abrupt collapse and re-establishment of turbulence in the SBL.

- Connect the fold geometry and slow passage phenomena to spurious finite-time modal blowup in low-order Galerkin approximations, identifying parameter regimes and truncation choices that amplify or suppress such pathologies.

4 Expected contributions

The thesis will provide:

- A mathematically rigorous fast–slow analysis of a prototypical SBL closure with regularization.
- A classification of critical manifold branches, folds, and hysteresis mechanisms relevant to turbulence collapse and recovery.
- A diagnostic framework for detecting and mitigating spurious finite-time blowup in low-order Galerkin and single-column boundary-layer models.